CGE-P LAB NOTES

# Lab 2.4: Terraform Modules for Compliance (GCP)

Lab 2.3 built one bucket. This lab builds a pattern. The shift to feel: you don't deploy buckets, you deploy a module that deploys buckets, and the security floor sits inside the module where consumers cannot reach it.

It is on GCP on purpose. Doing the same controls in a second cloud is what proves the control IDs are portable and the resource types are not.

For reading. Code lines wrap to fit the page; copy code from docs/02_04_terraform_modules_for_compliance.md, not the PDF.

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FROM CHECKBOX GRC TO ENGINEERED SYSTEMS

**This PDF is for reading.** Long code lines wrap to fit the page, so copy commands and code from `docs/02_04_terraform_modules_for_compliance.md` in your clone, not from the PDF.

## Learning objectives

- Compose a module with a clear interface: inputs, outputs, and hardcoded compliance defaults.
- Encode SC-8, SC-12, SC-13, SC-28, AU-3, AU-11, AC-3, and CM-6 so a consumer cannot switch them off.
- Distinguish a control you **implement** from one you **inherit** from the platform, and express the difference honestly.
- Emit a compliance attestation that Chapter 3 asserts against and Chapter 6 cites.

## Controls implemented

| Control | Enforced by | Note |
|---|---|---|
| SC-12 | `google_kms_crypto_key.rotation_period` | 90-day rotation. |
| SC-13, SC-28 | `encryption.default_kms_key_name` (CMEK) | Your key, not Google's. |
| AC-3 | `uniform_bucket_level_access` + `public_access_prevention = "enforced"` | |
| AU-3 | `logging.log_bucket` | Logs go somewhere other than the bucket being logged. |
| AU-9 | Versioning on the log bucket | |
| AU-11 | `retention_policy` + `lifecycle_rule` | Two different mechanisms; see Step 4. |
| CM-6, CM-8 | Merged required labels | Consumers may add, never suppress. |
| **SC-8** | **Inherited from GCP** | See below. This is the interesting one. |

### The inherited control

On AWS you write a bucket policy denying `aws:SecureTransport = false`, because S3 will happily serve plain HTTP if you let it. On GCP there is no equivalent resource, because the Cloud Storage JSON and XML APIs are TLS-only. The control is satisfied, and you did not implement it.

That distinction has a name in OSCAL: an `implementation-status` of `inherited`, with the provider named as the responsible party. It is not a loophole and it is not a gap. It is the correct answer, and writing "implemented" where you mean "inherited" is the kind of small dishonesty that collapses an assessment when someone asks you to show the code.

Lab 6.1 encodes this. Keep it in mind while you write the module.

## Prerequisites

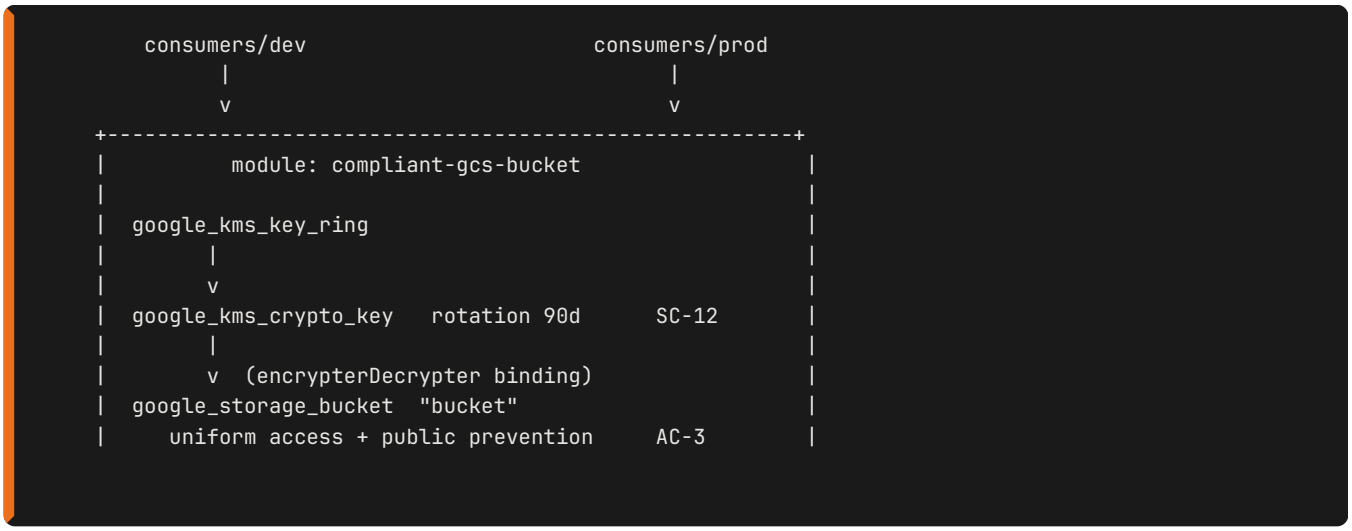
- Run these from inside the devcontainer if you set one up in Lab 0.1: that is where the toolchain and your cloud logins live. `source cgep.env` first, in every new shell.
- **Lab 0.1 Part 2** for the GCP side: creating the project, attaching billing, enabling the APIs, and the two separate authentications. This is the first GCP lab, so do that first if you have only set up AWS so far.
- A GCP project you control with billing enabled. Examples use `your-gcp-project`.
- `gcloud auth login` **and** `gcloud auth application-default login`. The google provider uses ADC.
- Roles: `roles/storage.admin`, `roles/cloudkms.admin`.
- `gcloud services enable cloudkms.googleapis.com`.
- Terraform `>= 1.9`. The cross-variable validation in Step 3 needs it.

## Estimated time & cost

- Time: 60 to 75 minutes.
- Cost: KMS key versions bill about **$0.06 per active version per month**, and 90-day rotation means versions accumulate. Storage is free while the buckets are empty. Destroy same-day and the prorated cost is fractions of a cent.

Read the Cleanup section before you apply. GCP KMS keys cannot be truly deleted, and one setting in this module can make a bucket undestroyable for a year.

## Architecture



```
    |     CMEK                                  SC-13/SC-28 |
    |     versioning + retention_policy         AU-11       |
    |     lifecycle_rule (noncurrent expiry)    AU-11       |
    |     logging -> log bucket                 AU-3        |
    |     merged required labels                CM-6/CM-8   |
    |                                                       |
    |  google_storage_bucket  "log"                         |
    |     versioning                            AU-9        |
    |     uniform access + public prevention    AC-3        |
    +-------------------------------------------------------+
```

Two buckets now, not one. The access-log bucket is an addition here, and it exists for the same reason it exists in Lab 2.3.

## Step-by-step walkthrough

### Concept: designing the interface

A module is a directory with an interface. The body decides what is hardcoded; the interface decides what consumers can change. Three rules:

1. `main.tf` hardcodes anything compliance-relevant: encryption, uniform access, versioning, logging, required labels.
2. `variables.tf` exposes only what the business actually changes: project, environment, retention duration, names.
3. `outputs.tf` returns evidence, including a computed attestation.

**Notice what the module does not contain: a `provider` block.** Providers are configured by the root module and inherited by children. This is the single most important structural difference between this lab and Lab 2.3, and Lab 2.3 gets it wrong on purpose. Lab 2.3's "primitive" declares its own provider, which makes it a root module wearing the word "primitive." The moment you try to call it twice for two regions, that provider block is what stops you.

If you take one thing from this lab into your capstone, take that.

### Step 1 `modules/compliant-gcs-bucket/main.tf`

```
# main.tf
terraform {
  required_version = ">= 1.9"
  required_providers {
    google = { source = "hashicorp/google", version = "~> 5.0" }
  }
  # No provider block. Consumers configure it; this module inherits it.
}
```

```
locals {
  # CM-6 / CM-8: the four required labels. Merged ON TOP of consumer labels,
  # which is the asymmetry that makes them non-negotiable.
  required_labels = {
    project          = var.project_label
    environment      = var.environment
    managed_by       = "terraform"
    compliance_scope = "cge-p-lab"
  }

  effective_labels = merge(var.labels, local.required_labels)

  bucket_name = "${var.project_label}-${var.environment}-${var.bucket_name_suffix}"
  log_name    = "${var.project_label}-${var.environment}-${var.bucket_name_suffix}-logs"
  keyring_id  = "${var.bucket_name_suffix}-ring"
  key_id      = "${var.bucket_name_suffix}-key"
}

data "google_storage_project_service_account" "gcs" {
  project = var.gcp_project
}

# SC-12: cryptographic key establishment and management. We own the key.
resource "google_kms_key_ring" "ring" {
  name     = local.keyring_id
  location = var.kms_location
  project  = var.gcp_project
}

# SC-12 / SC-13: 90-day rotation, expressed in seconds because the API is
# not sorry about it.
resource "google_kms_crypto_key" "key" {
  name            = local.key_id
  key_ring        = google_kms_key_ring.ring.id
  rotation_period = "7776000s" # 90 days

  lifecycle {
    prevent_destroy = false # set true in production
  }
}

resource "google_kms_crypto_key_iam_member" "gcs_encrypter" {
  crypto_key_id = google_kms_crypto_key.key.id
  role          = "roles/cloudkms.cryptoKeyEncrypterDecrypter"
  member        = "serviceAccount:${data.google_storage_project_service_account.gcs.email_address}"
}

# ------------------------------------------------------------------------
# AU-3 / AU-9: the access-log bucket. Logs belong somewhere other than the
# thing they are logging.
# ------------------------------------------------------------------------
resource "google_storage_bucket" "log" {
  name     = local.log_name
  project  = var.gcp_project
  location = var.location
```

```
  uniform_bucket_level_access = true
  public_access_prevention    = "enforced"

  # AU-9: audit records must survive overwrite.
  versioning {
    enabled = true
  }

  # AU-11: logs expire on a schedule instead of accumulating forever.
  lifecycle_rule {
    condition {
      age = var.log_retention_days
    }
    action {
      type = "Delete"
    }
  }

  labels = local.effective_labels
}

# GCS writes access logs as the Cloud Storage Analytics group. This binding is
# the GCP analogue of the AWS log-delivery bucket-policy grant in Lab 2.3.
resource "google_storage_bucket_iam_member" "log_writer" {
  bucket = google_storage_bucket.log.name
  role   = "roles/storage.objectCreator"
  member = "group:cloud-storage-analytics@google.com"
}

# ------------------------------------------------------------------------------
# The primary bucket. AC-3 + SC-28 + AU-3 + AU-11 + CM-6 in one declaration.
# ------------------------------------------------------------------------------
resource "google_storage_bucket" "bucket" {
  name     = local.bucket_name
  project  = var.gcp_project
  location = var.location

  # AC-3: uniform access removes per-object ACLs entirely (the GCP equivalent
  # of BucketOwnerEnforced), and public_access_prevention refuses public
  # grants even if someone with IAM rights tries to add one.
  uniform_bucket_level_access = true
  public_access_prevention    = "enforced"

  versioning {
    enabled = true
  }

  # SC-13 / SC-28: CMEK. Without this block GCS still encrypts at rest with a
  # Google-managed key, which is encryption you cannot rotate, scope, or audit.
  encryption {
    default_kms_key_name = google_kms_crypto_key.key.id
  }

  # AU-11, mechanism one: a retention policy is WORM. Objects cannot be
  # deleted until they reach this age. See Step 4 before setting is_locked.
```

```
  retention_policy {
    retention_period = var.retention_days * 86400
    is_locked        = var.lock_retention_policy
  }

  # AU-11, mechanism two: lifecycle expires superseded versions. Versioning
  # without this is an unbounded bill.
  lifecycle_rule {
    condition {
      num_newer_versions = var.keep_noncurrent_versions
      with_state         = "ARCHIVED"
    }
    action {
      type = "Delete"
    }
  }

  # AU-3: access logs to the dedicated bucket.
  logging {
    log_bucket        = google_storage_bucket.log.name
    log_object_prefix = "access-logs/"
  }

  labels = local.effective_labels

  depends_on = [google_kms_crypto_key_iam_member.gcs_encrypter]
}
```

The `depends_on` is genuine. The GCS service account must hold encrypt/decrypt on the key **before** the bucket is created, and there is no data dependency between the bucket and the IAM binding for Terraform to infer it from. Same shape as the legitimate `depends_on` in Lab 2.3.

## Step 2 The two-mechanism retention lesson

`retention_policy` and `lifecycle_rule` both appear above, and students routinely think one is redundant. They do opposite jobs:

| | `retention_policy` | `lifecycle_rule` |
|---|---|---|
| Effect | Objects **cannot be deleted** before this age | Objects **are** deleted at this age |
| Control | AU-9, AU-11 (preservation) | AU-11 (retention limit), cost |
| Direction | A floor | A ceiling |
| Reversible | Only if `is_locked = false` | Yes |

A compliance regime usually wants both: keep for at least N (you cannot destroy evidence early), delete after M (you do not hold data longer than the policy allows). Holding data past its retention limit is itself a finding under most privacy regimes.

## Step 3 `variables.tf` with validation

```
# variables.tf
variable "gcp_project" {
  type        = string
  description = "GCP project ID where the bucket and KMS resources live."
}

variable "location" {
  type        = string
  description = "GCS bucket location. Multi-regions like US and EU are valid."
  default     = "us-central1"
}

variable "kms_location" {
  type        = string
  description = "KMS keyring location. Must be a single region; multi-regions are not supported."
  default     = "us-central1"
}

variable "project_label" {
  type        = string
  description = "Short project identifier."
  validation {
    condition     = can(regex("^[a-z][a-z0-9-]{2,20}$", var.project_label))
    error_message = "project_label must be 3-21 lowercase alphanumerics or hyphens, starting with a
letter."
  }
}

variable "environment" {
  type        = string
  description = "Deployment environment."
  validation {
    condition     = contains(["dev", "staging", "prod"], var.environment)
    error_message = "environment must be one of: dev, staging, prod."
  }
}

variable "retention_days" {
  type        = number
  description = "Minimum object retention in days. Production must be >= 365."

  validation {
    condition     = var.retention_days >= 1 && var.retention_days <= 3650
    error_message = "retention_days must be between 1 and 3650."
  }

  validation {
    condition     = var.environment != "prod" || var.retention_days >= 365
    error_message = "retention_days must be >= 365 when environment == \"prod\"."
  }
}

variable "log_retention_days" {
```

```
  type        = number
  description = "AU-11. Age at which access-log objects are deleted."
  default     = 90
}

variable "keep_noncurrent_versions" {
  type        = number
  description = "How many superseded versions to retain before lifecycle deletes them."
  default     = 3
}

variable "lock_retention_policy" {
  type        = bool
  description = "IRREVERSIBLE. Locking a retention policy prevents shortening or removing it, and the
bucket cannot be deleted until every object ages out."
  default     = false
}

variable "bucket_name_suffix" {
  type        = string
  description = "Globally-unique suffix appended to the bucket name."
  validation {
    condition     = can(regex("^[a-z0-9-]{3,30}$", var.bucket_name_suffix))
    error_message = "bucket_name_suffix must be 3-30 lowercase alphanumerics or hyphens."
  }
}

variable "labels" {
  type        = map(string)
  description = "Optional additional labels. Required compliance labels merge on top."
  default     = {}
}
```

**Why two location variables.** GCS buckets accept multi-region names like `US` and `EU`. KMS keyrings do not. Setting both from one variable fails with `KMS_RESOURCE_NOT_FOUND_IN_LOCATION` the first time you try `US`. Splitting them keeps the lesson honest.

**`lock_retention_policy` is an input, not a hardcoded `false`.** An irreversible choice belongs in the consumer's code where a reviewer sees it, rather than buried in a module the consumer never opens. The loud description is part of the control.

## Step 4 `outputs.tf` as evidence

```
# outputs.tf
output "bucket_url" {
  value       = google_storage_bucket.bucket.url
  description = "gs:// URL of the compliant bucket."
```

```
}

output "bucket_name" {
  value       = google_storage_bucket.bucket.name
  description = "Name of the compliant bucket."
}

output "log_bucket_name" {
  value       = google_storage_bucket.log.name
  description = "Name of the access-log bucket."
}

output "kms_key_id" {
  value       = google_kms_crypto_key.key.id
  description = "Resource ID of the CMEK protecting this bucket."
}

output "compliance_attestation" {
  description = "Computed attestation of the controls this module enforces."
  value = {
    encryption_type          = "cmek"
    kms_key_id               = google_kms_crypto_key.key.id
    kms_rotation_period      = google_kms_crypto_key.key.rotation_period
    versioning_enabled       = google_storage_bucket.bucket.versioning[0].enabled
    log_versioning_enabled   = google_storage_bucket.log.versioning[0].enabled
    public_access_prevention = google_storage_bucket.bucket.public_access_prevention
    uniform_access_enforced  = google_storage_bucket.bucket.uniform_bucket_level_access
    access_logging_target    = google_storage_bucket.bucket.logging[0].log_bucket
    retention_period_days    = var.retention_days
    retention_policy_locked  = var.lock_retention_policy
    log_retention_days       = var.log_retention_days

    required_labels_present = alltrue([
      for k in keys(local.required_labels) :
      contains(keys(google_storage_bucket.bucket.labels), k)
    ])

    # SC-8 is satisfied by the platform, not by this module. Saying so in the
    # attestation is the honest form; Lab 6.1 maps this to an OSCAL
    # implementation-status of "inherited".
    transit_encryption = "inherited-gcp-tls-only"
  }
}
```

Compare this shape to Lab 2.3's `compliance_attestation`. Different cloud, different resource types, same evidence surface. That similarity is what lets one Rego library and one OSCAL component describe both.

## Step 5 Consumer: dev

```
# consumers/dev/main.tf
terraform {
  required_version = ">= 1.9"
  required_providers {
    google = { source = "hashicorp/google", version = "~> 5.0" }
  }
}

provider "google" {
  project = var.gcp_project
  region  = "us-central1"
}

variable "gcp_project" { type = string }

module "data_bucket" {
  source = "../../modules/compliant-gcs-bucket"

  gcp_project        = var.gcp_project
  project_label      = "cgep-lab"
  environment        = "dev"
  retention_days     = 30
  bucket_name_suffix = "dev-data-001"
}

output "attestation" { value = module.data_bucket.compliance_attestation }
output "bucket_url"  { value = module.data_bucket.bucket_url }
```

Six lines of business configuration. Twenty-plus controls. The provider lives here, in the root module, exactly where it belongs.

## Step 6 Consumer: prod

Same module, two values changed:

```
module "data_bucket" {
  source = "../../modules/compliant-gcs-bucket"

  gcp_project        = var.gcp_project
  project_label      = "cgep-lab"
  environment        = "prod"
  retention_days     = 365
  bucket_name_suffix = "prod-data-001"
}
```

## Step 7 Apply dev

Plan prod, but do not apply it. A 365-day retention takes a year to expire.

```
cd consumers/dev
terraform init
terraform fmt && terraform validate
terraform plan -out=tfplan -var=gcp_project=your-gcp-project
terraform apply -auto-approve tfplan
```

Tail:

```
attestation = {
  "access_logging_target"    = "cgep-lab-dev-dev-data-001-logs"
  "encryption_type"          = "cmek"
  "kms_rotation_period"      = "7776000s"
  "log_retention_days"       = 90
  "log_versioning_enabled"   = true
  "public_access_prevention" = "enforced"
  "required_labels_present"  = true
  "retention_period_days"    = 30
  "retention_policy_locked"  = false
  "transit_encryption"       = "inherited-gcp-tls-only"
  "uniform_access_enforced"  = true
  "versioning_enabled"       = true
}
```

That output is your SC-12 / SC-13 / SC-28 / AC-3 / AU-3 / AU-9 / AU-11 / CM-6 attestation in machine-readable form.

## Step 8 The negative test

Copy `consumers/dev` to `consumers/negative-test`, set `environment = "prod"` and leave `retention_days = 30`:

```
Error: Invalid value for variable

  var.environment is "prod"
  var.retention_days is 30

retention_days must be >= 365 when environment == "prod".

This was checked by the validation rule at variables.tf:...
```

This is the lesson. The compliance check ran at `terraform plan`, before any resource existed, with a message specific enough that the developer fixes it without filing a ticket. Preventive beats detective, and it is cheaper.

Run a second negative test: try to suppress a required label.

```
labels = { compliance_scope = "not-in-scope" }
```

Plan it. The label comes back as `cge-p-lab`, because `merge(var.labels, local.required_labels)` puts the required set last and last wins. **The consumer can add labels and cannot override the compliance ones.** That asymmetry is the entire design, and watching it refuse is better than reading about it.

## Verification

```
gcloud storage buckets describe gs://cgep-lab-dev-dev-data-001 \
  --format="yaml(uniform_bucket_level_access,public_access_prevention,labels,retention_policy,logging)
"

gcloud storage buckets describe gs://cgep-lab-dev-dev-data-001 \
  --format="value(default_kms_key,versioning_enabled)"

gcloud kms keys describe dev-data-001-key \
  --keyring=dev-data-001-ring --location=us-central1 \
  --format="value(rotationPeriod,nextRotationTime)"

# AU-9: the log bucket is versioned too
gcloud storage buckets describe gs://cgep-lab-dev-dev-data-001-logs \
  --format="value(versioning_enabled)"
```

### Capture the evidence the checklist asks for

```
# evidence/ lives at the repository root, not in the workspace you are in
EVIDENCE="$(git rev-parse --show-toplevel)/evidence/lab-2-4"
mkdir -p "$EVIDENCE"
terraform show -json tfplan > "$EVIDENCE/plan.json"
terraform output -json compliance_attestation > "$EVIDENCE/attestation.json"
```

The negative test is the one worth keeping, because it is the proof the module refuses rather than merely defaults:

```
( cd ../negative-test && terraform plan 2>&1 ) | tee "$EVIDENCE/negative-test.txt"
```

The subshell is deliberate: the `cd` ends with it, so the `tee` path stays relative to where you started and you are not left in a different directory.

## Portfolio submission checklist

- [ ] `terraform/modules/compliant-gcs-bucket/{main.tf,variables.tf,outputs.tf,README.md}`
- [ ] `terraform/consumers/{dev,prod}/` committed.
- [ ] Module README lists each control by family **and marks SC-8 as inherited, with the reason.**
- [ ] `evidence/lab-2-4/plan.json`
- [ ] `evidence/lab-2-4/attestation.json` from `terraform output -json compliance_attestation`
- [ ] `evidence/lab-2-4/negative-test.txt` capturing both refusals: the prod retention rule and the label-override attempt.

## Troubleshooting

- `KMS_RESOURCE_NOT_FOUND_IN_LOCATION` when `location` is `US`. Keyrings need a single region. The two-variable split is the fix.
- `Permission cloudkms.cryptoKeyEncrypterDecrypter denied` during bucket creation. The GCS service agent needs encrypt/decrypt on the key before the bucket exists. The `depends_on` sequences it; keep it if you refactor.
- **Access logs never appear.** GCS access logs are delivered roughly hourly, and only when there is traffic. Confirm the `cloud-storage-analytics@google.com` binding exists on the log bucket.
- **Retention policy cannot be shortened.** It can be lengthened or removed only while `is_locked = false`. Once locked, neither.
- `googleapi: Error 409: ... already exists`. GCS names are globally unique. Change `bucket_name_suffix`.
- `reauth related error (invalid_rapt)`. Run `gcloud auth application-default login` again. ADC tokens expire and Terraform will not refresh them.

## Cleanup

```
cd consumers/dev
terraform destroy -auto-approve
```

Three things that will surprise you:

1. **A locked retention policy makes the bucket undestroyable** until every object ages out. With `retention_days = 365` and `lock_retention_policy = true`, that is a year. This is why the variable defaults to `false` and says so loudly.
2. **KMS crypto keys are never truly deleted.** Destroying a key version enters a 30-day soft-delete. The key and keyring objects remain indefinitely; keyrings cannot be deleted at all. They are free, so this is tidiness rather than cost, but a `terraform destroy` that reports success has not removed them.
3. **The log bucket may hold objects** and refuse to destroy. Empty it first if the bucket saw traffic.

## How this feeds the capstone

- **The module discipline is the deliverable**, more than the module. Your capstone's KMS and S3 hardening should be a module both dev and prod consume, so the floor is enforced once.
- **Ch 3**, Rego reads `compliance_attestation` and refuses to merge a plan that cannot produce it.
- **Ch 6**, the OSCAL component for `compliant-gcs-bucket-v1` cites this module path as its implementation and the attestation JSON as evidence. The `transit_encryption` field becomes an `inherited` implementation status, which is the honest OSCAL and the thing most candidates get wrong.

# Revision history

### These notes

- Added an access-log bucket, with versioning and a lifecycle rule, adding AU-3 and AU-9. The module previously logged nowhere.
- `lock_retention_policy` exposed as an input rather than hardcoded, so an irreversible choice appears in the consumer's code.
- Added a second lifecycle mechanism for superseded versions, and documented why `retention_policy` (a floor) and `lifecycle_rule` (a ceiling) are both needed.
- The attestation now records `transit_encryption` as inherited from the platform, rather than leaving SC-8 unstated.
- Added a second negative test showing that a consumer cannot suppress a required label.

### The official labs

Initial release: single bucket, no access logging, `is_locked` hardcoded.